BT403 - Agriculture Biotechnology

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Assignment No. 02

Spring 2024
Due Date: July 03, 2024

Assignment Topic:

Imagine a plant is facing any type of biotic stress. How the plant will manage to survive under stress condition. Take any example of your choice (any biotic stress) and explain the defence mechanism that plant will adopt against that biotic stress.
Note: You can choose any biotic stress factor e.g. Bacteria, Virus, Fungus, Nematodes, Insect-pests etc. Your answer should be according to chosen stress factor.

Plant Defense Mechanisms Against Fungal Stress: Wheat vs. Fusarium graminearum

Introduction

Plants, as sessile organisms, have evolved sophisticated mechanisms to defend themselves against various biotic stresses, including fungal pathogens. In this essay, we will explore how plants, specifically wheat (Triticum aestivum), manage to survive under stress conditions caused by the fungal pathogen Fusarium graminearum, the causative agent of Fusarium Head Blight (FHB).

Overview of Fusarium Head Blight (FHB)

Fusarium Head Blight, also known as scab, is a devastating disease that affects wheat and other small grain cereals worldwide. The disease is caused by several species of Fusarium, with F. graminearum being the most prevalent. FHB not only reduces yield but

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also contaminates grains with mycotoxins, particularly deoxynivalenol (DON), which poses serious health risks to humans and animals.

Plant Defense Mechanisms Against Fusarium graminearum

When faced with F. graminearum infection, wheat plants employ a multi-layered defense strategy that includes both constitutive and induced responses. These defense mechanisms can be broadly categorized into physical, chemical, and molecular responses.

1. Physical Barriers

The first line of defense against F. graminearum is the plant's physical barriers:

a) Cuticle and Cell Wall: The cuticle, a waxy layer on the plant surface, and the cell wall act as physical barriers against fungal penetration. Wheat plants may reinforce these structures upon pathogen detection by depositing additional layers of cutin, suberin, and lignin.

b) Stomatal Closure: Although F. graminearum primarily infects through the flowers (anthers), wheat plants can regulate stomatal apertures to limit fungal entry through these natural openings.

2. Chemical Defenses

If the fungus breaches the physical barriers, wheat plants activate various chemical defenses:

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a) Antimicrobial Compounds: Wheat produces an array of antimicrobial compounds, including phenolic compounds, flavonoids, and terpenes. For instance, the phenylpropanoid pathway is activated, leading to the production of lignin precursors and other phenolic compounds with antifungal properties.

b) Phytoalexins: These are low molecular weight antimicrobial compounds synthesized de novo in response to pathogen attack. In wheat, benzoxazinoids like DIMBOA (2,4-dihydroxy-7-methoxy-1,4-benzoxazin-3-one) have been shown to have antifungal activity against F. graminearum.

c) Pathogenesis-Related (PR) Proteins: Wheat plants produce various PR proteins with antifungal activities, such as chitinases (PR-3, PR-4, PR-8, PR-11) that degrade fungal cell walls, and β-1,3-glucanases (PR-2) that hydrolyze β-1,3-glucans in fungal cell walls.

3. Molecular and Cellular Responses

The plant's defense response is orchestrated by complex molecular and cellular mechanisms:

a) Pathogen Recognition: Wheat plants recognize Pathogen-Associated Molecular Patterns (PAMPs) of F. graminearum using Pattern Recognition Receptors (PRRs). This recognition triggers PAMP-Triggered Immunity (PTI), the first layer of induced defense.

b) Signaling Cascades: Upon pathogen recognition, various signaling cascades are activated, involving:

- Mitogen-Activated Protein Kinase (MAPK) cascades
- Calcium signaling
- Reactive Oxygen Species (ROS) production
- Phytohormone signaling (particularly salicylic acid, jasmonic acid, and ethylene)

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c) Transcriptional Reprogramming: These signaling events lead to extensive transcriptional reprogramming, activating numerous defense-related genes.

d) Hypersensitive Response (HR): In some cases, wheat may initiate localized cell death (HR) to contain the fungal infection. However, as F. graminearum is a hemibiotrophic pathogen with both biotrophic and necrotrophic phases, the effectiveness of HR can vary depending on the infection stage.

4. Specific Defense Mechanisms Against F. graminearum

a) Detoxification of Mycotoxins: F. graminearum produces the mycotoxin deoxynivalenol (DON) as a virulence factor. Wheat plants have evolved mechanisms to detoxify DON, including:

- UDP-glycosyltransferases that convert DON to less toxic DON-3-glucoside
- ABC transporters that may be involved in DON efflux

b) Resistance Genes: Several quantitative trait loci (QTLs) contributing to FHB resistance have been identified in wheat, including Fhb1, Fhb2, and Fhb5. These genes enhance various aspects of resistance, such as initial infection resistance, spread resistance, and DON accumulation resistance.

c) Modification of Cell Wall Composition: Wheat plants may alter their cell wall composition to enhance resistance. For example, increased deposition of hydroxycinnamic acid amides (HCAAs) in cell walls has been associated with improved resistance to F. graminearum.

Temporal and Spatial Aspects of Defense Response

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The defense response of wheat against F. graminearum is not uniform but varies temporally and spatially:

1. Temporal Variation:

- Early responses (within hours): ROS production, MAPK activation, early defense gene induction
- Intermediate responses (within days): PR protein accumulation, phytoalexin production
- Late responses (several days post-infection): systemic acquired resistance, long-term transcriptional changes

2. Spatial Variation:

- Local responses: Occur at the site of infection, including HR and local defense gene activation
- Systemic responses: Activation of defenses in uninfected parts of the plant, preparing for potential future infections

Challenges and Adaptations

Despite these defense mechanisms, F. graminearum often successfully infects wheat. This is due to several factors:

1. Fungal Counter-strategies: F. graminearum has evolved mechanisms to suppress or evade plant defenses, such as secreting effector proteins that interfere with plant immunity.

2. Environmental Factors: The effectiveness of plant defenses can be influenced by environmental conditions. For instance, high humidity favors fungal growth and can overwhelm plant defenses.

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3. Genetic Variation: Different wheat varieties exhibit varying levels of resistance to FHB, reflecting differences in their defense capabilities.

To address these challenges, ongoing research focuses on:

1. Breeding for Resistance: Developing wheat varieties with improved FHB resistance by incorporating resistance genes from diverse wheat germplasm and related species.

2. Genetic Engineering: Creating transgenic wheat with enhanced defense capabilities, such as overexpressing genes involved in DON detoxification.

3. Integrated Disease Management: Combining genetic resistance with cultural practices and fungicide applications to manage FHB effectively.

Conclusion

The defense mechanisms employed by wheat against Fusarium graminearum illustrate the complex and multi-faceted nature of plant immunity. From preformed physical barriers to induced chemical defenses and intricate signaling networks, plants have evolved a sophisticated array of strategies to cope with fungal pathogens. Understanding these mechanisms not only provides insights into plant-pathogen interactions but also informs efforts to develop more resistant crop varieties and sustainable disease management strategies.

As research in this field continues, new layers of complexity in plant defense responses are being uncovered, including the role of small RNAs, epigenetic regulation, and the plant microbiome in modulating resistance. These advances promise to further our

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understanding of plant immunity and may lead to novel approaches for enhancing crop resistance to devastating diseases like Fusarium Head Blight.

General instruction:

Dear Students:
Please note this assignment is "Inline" assignment and copy paste is not allowed. You will write your answers within given space.

- The length of assignment should be 700-1000 words.
- Give the title at top of the assignment with proper subheadings.
- Please be specific and to the point and give the answer according to question avoiding irrelevant details.
- This is not a group assignment; it is an individual assignment so be careful and avoid copying others' work.
- Give references where necessary.

- This assignment carries 10 marks.
- Assignment should be submitted on LMS not later than **July 03, 2024.**

DO NOT send the assignment thorough email. Assignments submitted through email will result in no marks.

Outcomes:

This assignment will help you to understand the plant's response against biotic stresses.

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